IIE6120 Linear Programming
2025-2 Homework 01
Team B (이정민, 장서현)

#1

1. Formulate the production planning model discussed in class as an LP with only X_t variables.

Production Planning : Multi-period inv

Set and indices:

- Discretized periods: $t = 1, \ldots, T$

Parameters:

- c_P: cost of producing one unit of the product
- c_I: cost of carrying one unit of the product in inventory from period t to $t+1$: holding cost (단가×이자율/year/52)
- d_t: demand of period t
- u_t: production capacity in period t
- I_0: initial inventory

Variables:

- X_t: number of product units produced in period t

LP Formulation

$$\min \sum_{t=1}^{T} c_P X_t + \sum_{t=1}^{T} c_I \left(\sum_{k=1}^{t} (X_k - d_k) + I_0 \right)$$

$$\text{s.t.} \quad 0 \leq X_t \leq u_t \quad \forall t = 1, \cdots, T$$

$$I_0 + \sum_{k=1}^{t} (X_k - d_k) \geq 0 \quad \forall t = 1, \cdots, T$$

+) index k : dummy index for summation ($k = 1, \cdots, t$)

#2

[1.1] Fred has $5000 to invest over the next five years. At the beginning of each year he can invest money in one– or two–year time deposits. The bank pays 4 percent interest on one–year time deposits and 9 percent (total) on two–year time deposits. In addition, West World Limited will offer three–year certificates starting at the beginning of the second year. These certificates will return 15 percent (total). If Fred reinvests his money that is available every year, formulate a linear program to show him how to maximize his total cash on hand at the end of the fifth year.

DV

- x_t : Amount invested in one-year deposits at beginning of year t ($t = 1, 2, 3, 4, 5$)
- y_t : Amount invested in two-year deposits at beginning of year t ($t = 1, 2, 3, 4$)
- z : Amount invested in three-year certificate at beginning of year 2.
- S_t : Amount of cash saved at beginning of year t. ($t = 1, 2, 3, 4, 5$)

LP Formulation

$$\max \; 1.04x_5 + 1.09y_4 + 1.15z + S_5$$

$$\begin{aligned}
\text{s.t.} \quad & x_1 + y_1 + S_1 = 5000 \\
& x_2 + y_2 + z + S_2 = 1.04x_1 + S_1 \\
& x_3 + y_3 + S_3 = 1.04x_2 + 1.09y_1 + S_2 \\
& x_4 + y_4 + S_4 = 1.04x_3 + 1.09y_2 + S_3 \\
& x_5 + S_5 = 1.04x_4 + 1.09y_3 + S_4 \\
& x_t, y_t, z, S_t \geq 0 \quad \forall t
\end{aligned}$$

#3 **[1.17]** A 10–acre slum in New York City is to be cleared. The officials of the city must decide on the redevelopment plan. Two housing plans are to be considered: low–income housing and middle–income housing. These types of housing can be developed at 20 and 15 units per acre, respectively. The unit costs of the low– and middle–income housing are \$17,000 and \$25,000. The lower and upper limits set by the officials on the number of low–income housing units are 80 and 120. Similarly, the number of middle–income housing units must lie between 40 and 90. The combined maximum housing market potential is estimated to be 190 (which is less than the sum of the individual market limits due to the overlap between the two markets). The total mortgage committed to the renewal plan is not to exceed \$2.5 million. Finally, it was suggested by the architectural adviser that the number of low–income housing units be at least 50 units greater than one–half the number of the middle–income housing units.

a. Formulate the minimum cost renewal planning problem as a linear program and solve it graphically.
b. Resolve the problem if the objective is to maximize the number of houses to be constructed.

a.

DV) x_1 = number of units by low-income housing

x_2 = number of units by middle-income housing

LP formulation)

$$\min\ z_1 = 17000x_1 + 25000x_2$$

$$\text{s.t.}\quad \frac{x_1}{20} + \frac{x_2}{15} \le 10$$

$$80 \le x_1 \le 120$$

$$40 \le x_2 \le 90$$

$$x_1 + x_2 \le 190$$

$$17000x_1 + 25000x_2 \le 2500000$$

$$x_1 \ge 0.5x_2 + 50$$

$\therefore (x_1, x_2) = (80, 40)$

b.

LP) $\max\ z_2 = x_1 + x_2$

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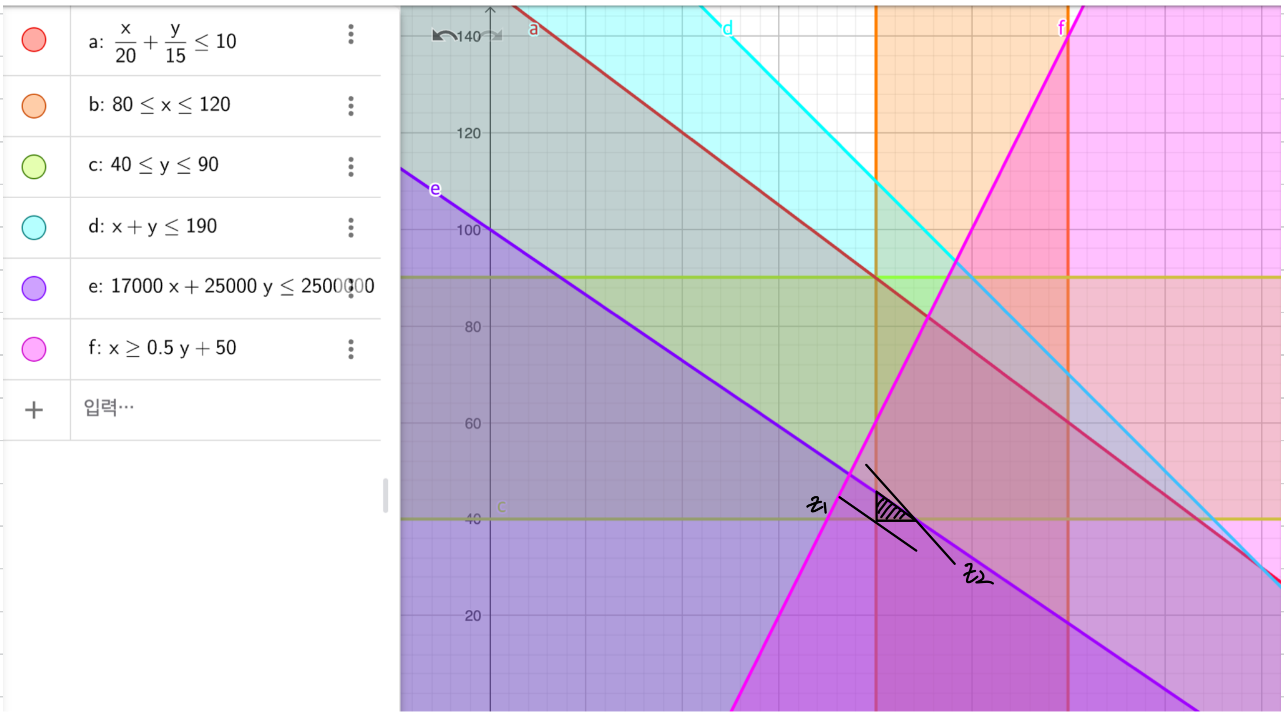
$(x_1, x_2) = (88, 40)$



#4 **[1.25]** Suppose that there are m sources that generate waste and n disposal sites. The amount of waste generated at source i is a_i and the capacity of site j is b_j. It is desired to select appropriate transfer facilities from among K candidate facilities. Potential transfer facility k has a fixed cost f_k, capacity q_k, and unit processing cost α_k per ton of waste. Let c_{ik} and $\overline{c}_{kj}$ be the unit shipping costs from source i to transfer station k and from transfer station k to disposal site j, respectively. The problem is to choose the transfer facilities and the shipping pattern that minimize the total capital and operating costs of the transfer stations plus the transportation costs. Formulate this *distribution problem*. (*Hint*: Let y_k be 1 if transfer station k is selected and 0 otherwise.)

DV) x_{ik}: amount of waste from source i to facility k

z_{kj}: amount of waste from facility k to disposal site j

index) $i=1,\dots,m$, $k=1,\dots,K$, $j=1,\dots,n$

LP) $$\min\ z = \sum_{i=1}^{m}\sum_{k=1}^{K} c_{ik}x_{ik} + \sum_{j=1}^{n}\sum_{k=1}^{K} \overline{c}_{kj}\cdot z_{kj} + \sum_{k=1}^{K} f_k\cdot y_k + \sum_{k=1}^{K}\sum_{i=1}^{m} \alpha_k\cdot x_{ik}$$

s.t. $\sum_{k=1}^{K} x_{ik} = a_i \quad \forall i$

$\sum_{k=1}^{K} z_{kj} \le b_j \quad \forall j$

$\sum_{i=1}^{m} x_{ik} = \sum_{j=1}^{n} z_{kj} \quad \forall k$

$\sum_{i=1}^{m} x_{ik} \le q_k\cdot y_k \quad \forall k$

$x_{ik}, z_{kj} \ge 0 \quad \forall i,k,j$

$y_k \in \{0,1\} \quad \forall k$

#5 **[1.36]** Consider the following problem:

$$\begin{array}{lrcrcrcrcl}
\text{Maximize} & -x_1 & - & x_2 & + & 2x_3 & + & x_4 & & \\
\text{subject to} & 2x_1 & + & x_2 & + & x_3 & + & x_4 & \ge & 6 \\
& x_1 & + & 2x_2 & - & 2x_3 & + & x_4 & \le & 4 \\
& x_1, & & x_2, & & x_3, & & x_4 & \ge & 0.
\end{array}$$

a. Introduce slack variables and draw the requirement space.
b. Interpret feasibility in the requirement space.
c. You are told that an optimal solution can be obtained by having at most two positive variables while all other variables are set at zero. Utilize this statement and the requirement space to find an optimal solution.

a. max $-x_1 - x_2 + 2x_3 + x_4$

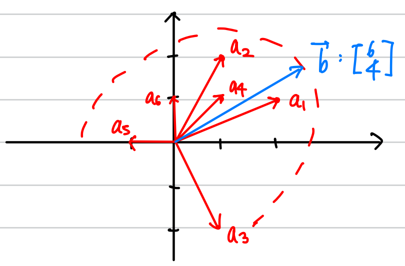
s.t $2x_1 + x_2 + x_3 + x_4 - S_1 = 6$

$x_1 + 2x_2 - 2x_3 + x_4 + S_2 = 4$

$x_1, x_2, x_3, x_4, S_1, S_2 \ge 0$

requirement space

$= \text{span}\left\{\begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}1\\-2\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}-1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$



b. $\vec{b} = \begin{bmatrix}6\\4\end{bmatrix}$ is included in the requirement space. Thus, feasible

c. When we can have at most two positive variables, the obj. function value reaches its maximum when x_3 and S_2 has positive values. $x_1 = x_2 = x_4 = S_1 = 0$

$x_3 = 6$, $S_2 = 16$, $z = 12$

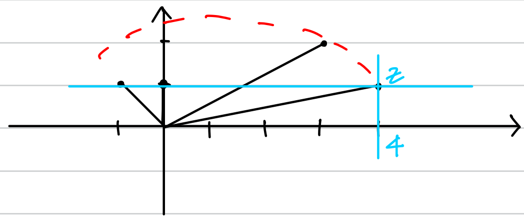
#6

6. Solve the following in the requirement space on a picture.

$$\begin{aligned} \text{max} \quad & z = -x_1 + 4x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + 2x_3 \le 1 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

$$\begin{aligned} \max \; & z = -x_1 + 4x_2 + 3x_3 \\ \text{s.t.} \; & x_1 + x_2 + 2x_3 + s_1 = 1 \\ & x_1, x_2, x_3, s_1 \ge 0. \end{aligned}$$

$$\Rightarrow \begin{bmatrix}-1\\1\end{bmatrix}x_1 + \begin{bmatrix}4\\1\end{bmatrix}x_2 + \begin{bmatrix}3\\2\end{bmatrix}x_3 + \begin{bmatrix}0\\1\end{bmatrix}s_1 = \begin{bmatrix}z\\1\end{bmatrix}$$



$\therefore z^* = 4$

$(x_1, x_2, x_3) = (0, 1, 0)$

#7

[1.37] Consider the following problem:

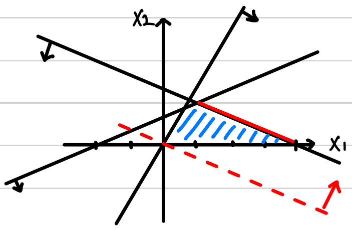
$$\begin{array}{lrcrcl} \text{Maximize} & 3x_1 & + & 6x_2 & & \\ \text{subject to} & -x_1 & + & 2x_2 & \le & 2 \\ & -2x_1 & + & x_2 & \le & 0 \\ & x_1 & + & 2x_2 & \le & 4 \\ & x_1, & & x_2 & \ge & 0. \end{array}$$

a. Graphically identify the set of all alternative optimal solutions to this problem.

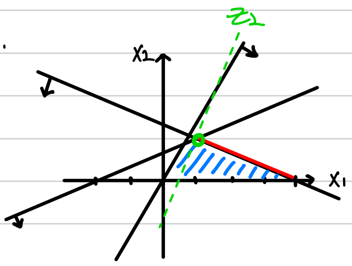
b. Suppose that a secondary priority objective function seeks to maximize $-3x_1 + x_2$ over the set of alternative optimal solutions identified in Part (a). What is the resulting solution obtained?

c. What is the solution obtained if the priorities of the foregoing two objective functions is reversed?

a. marked red : $\{(x_1, x_2) \mid x_1 + 2x_2 = 4,\ 1 \le x_1 \le 4,\ 0 \le x_2 \le \frac{3}{2}\}$

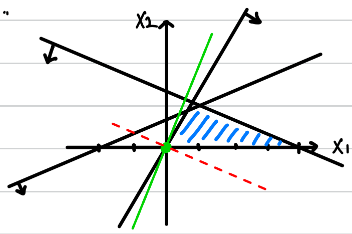


b.



$\therefore$ sol : $(x_1, x_2) = (1, \frac{3}{2})$

C.



$\therefore$ sol : $(x_1, x_2) = (0, 0)$

#8

[1.38] Consider the problem: Minimize **cx** subject to $\mathbf{Ax} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Suppose that one component of the vector **b**, say b_i, is increased by one unit to $b_i + 1$.

a. What happens to the feasible region?
b. What happens to the optimal objective value?

a. As b_i is increased by one unit, the feasible region is decreased or remains equal.
b. Minding that it is a minimization problem, as the feasible region decreases (or remains), the objective value increases (or remains).

[1.39] From the results of the previous problem, assuming $\partial z^* / \partial b_i$ exists, is it ≤ 0, $= 0$, or ≥ 0?

When b_i is increased by one unit, the objective value increase or remains equal, thus the answer is ≥.

[1.40] Solve Exercises 1.38 and 1.39 if the restrictions $\mathbf{Ax} \geq \mathbf{b}$ are replaced by $\mathbf{Ax} \leq \mathbf{b}$.

If the restrictions are replaced,
(38a) The feasible region is increased or remains equal.
(38b) Leadingly, the optimal objective value is decreased or remains equal.
(39) Thus, the answer is ≤.

[1.41] Consider the problem: Minimize **cx** subject to $\mathbf{Ax} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Suppose that a new constraint is added to the problem.

a. What happens to the feasible region?
b. What happens to the optimal objective value z^*?

a. The feasible region either shrinks or stays the same. Adding a new constraint restricts the set of possible solutions, so the new feasible region is a subset of the original one. It cannot get larger.
b. The optimal objective value z∗ either increases or stays the same. Since the problem is a minimization problem and the feasible region cannot grow, the minimum possible value of the objective function cannot decrease. If the original optimal solution remains feasible, the value won't change. If it becomes infeasible, the new minimum will be found at a point with a higher (or equal) objective value.

[1.42] Consider the problem: Minimize **cx** subject to $\mathbf{Ax} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Suppose that a new variable is added to the problem.

a. What happens to the feasible region?
b. What happens to the optimal objective value z^*?

a. The feasible region is projected into a higher-dimensional space, effectively enlarging the set of possible solutions.
b. The optimal objective value z∗either decreases or stays the same. Since the problem is a minimization problem and the set of possible solutions has expanded, a better (lower) objective value might be found. The value cannot increase because the original optimal solution is still a valid option (by setting the new variable to 0).

[1.43] Consider the problem: Minimize **cx** subject to $\mathbf{Ax} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Suppose that a constraint, say constraint i, is deleted from the problem.

a. What happens to the feasible region?
b. What happens to the optimal objective value z^*?

a. The feasible region either expands or stays the same. Removing a constraint means there are fewer conditions to satisfy; thus,it cannot get smaller.
b. The optimal objective value z∗ either decreases or stays the same. Since this is a minimization problem and the feasible region has expanded, a better (lower) objective value may become available.

[1.44] Consider the problem: Minimize **cx** subject to $\mathbf{Ax} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Suppose that a variable, say, x_k, is deleted from the problem.

a. What happens to the feasible region?
b. What happens to the optimal objective value z^*?

a. The feasible region shrinks or stays the same. The new feasible region is a projection of the original one onto a lower-dimensional space.
b. The optimal objective value z∗ either increases or stays the same. Since this is a minimization problem and the set of possible solutions has been restricted, the optimal value cannot decrease. If the original optimal solution had the deleted variable at zero, the value will not change; otherwise, it will increase.

#9

9. Consider the following LP:

$$\begin{aligned} \min \quad & z = c_1x_1 + c_2x_2 + c_3x_3 \\ \text{s.t.} \quad & x_2 \le 1 \\ & x_3 \le 1 \\ & x_1, x_2, x_3 \ge 0, \end{aligned}$$

By imagining the feasible region and objective plane in 3-dimensional space, for each of the following values for $c = (c_1, c_2, c_3)$, describe the set of optimal solutions (if any) in words and exact mathematical notation, and also write the optimal objective value : (i) $c = (-1, -1, -1)$; (ii) $c = (1, -1, -1)$; (iii) $c = (0, -1, -1)$; (iv) $c = (0, 0, -1)$.

(i) set of optimal solutions : $\emptyset$
optimal obj. value : Unbounded
(x_1 can increase infinitely)

(ii) set of optimal solutions : $\{(0, 1, 1)\}$
optimal obj. value : -2

(iii) set of optimal solutions : $\{(x_1, 1, 1) \mid x_1 \geq 0\}$
optimal obj. value : -2

(iv) set of optimal solutions : $\{(x_1, x_2, 1) \mid x_1 \geq 0,\ 0 \leq x_2 \leq 1\}$
optimal obj. value : -1